

RESULTS

Results

01 · INDEPENDENT T-TEST

do two groups' means differ?

Table 1. Group statistics

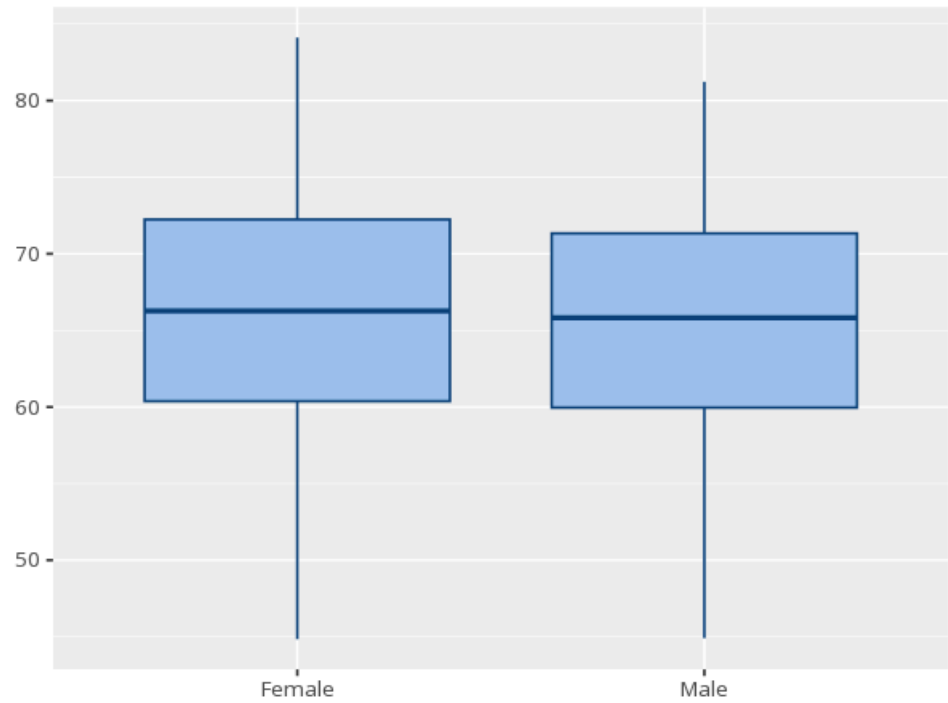
Group	N	Mean	Std. Dev.	SE
Female	120	65.69	8.31	0.76
Male	120	65.44	8.03	0.73

Table 2. Independent-samples t-test

Contrast	t	df	p	M _{diff}	95% CI	Cohen's d [95% CI]
Female – Male	0.24	237.73	.810	0.25	[-1.82, 2.33]	0.03 [-0.22, 0.28]

assumption checks: Levene's test for equal variances & within-group normality (Shapiro-Wilk per group); a Welch row replaces the pooled row when equal-variance is off. (Levene F=0.27, p=.605 · Shapiro Female W=0.99, p=.474; Shapiro Male W=0.99, p=.466 · Welch test) — equal variances look reasonable — normality looks reasonable in both groups

Figure. Distribution of the outcome by group



Understanding the numbers

t. How many standard errors the two group means are apart. Here, $t(237.73) = 0.24$.

p. The probability of a difference this large (or larger) if the groups truly had equal means. Here, $p = .810$ - at or above the conventional .05 threshold, no significant difference detected.

Cohen's d. The size of the gap in standard-deviation units (~ 0.2 small, 0.5 medium, 0.8 large). Here, $d = 0.03 [-0.22, 0.28]$.

N. The number of complete cases (rows) in each group. Here, $n = 120$ for Female and $n = 120$ for Male.

Mean. The average outcome value in each group. Here, $M = 65.69$ for Female and $M = 65.44$ for Male.

SD. How spread out the outcome values are around each group's mean. Here, $SD = 8.31$ for Female and $SD = 8.03$ for Male.

SE. The precision of each group's mean as an estimate of its population mean. Here, $SE = 0.76$ for Female and $SE = 0.73$ for Male.

df. The degrees of freedom for the t-test (pooled or Welch-adjusted). Here, $df = 237.73$.

M diff. The difference between the two group means. Here, the mean difference is 0.25.

95% CI. The range of mean differences plausible at the chosen confidence level, given this sample. Here, the 95% CI for the mean difference is $[-1.82, 2.33]$.

How to read this test

Asks whether two groups' averages differ by more than chance. Look at p: below alpha (e.g. 0.05) means significant. The mean difference and 95% CI show the size and precision of the gap; Cohen's d gives effect size (~ 0.2 small, 0.5 medium, 0.8 large). A large p means no significant difference was detected — not that the group means are equal.

APA template: An independent-samples t-test compared Female ($M=65.7$, $SD=8.3$) and Male ($M=65.4$, $SD=8.0$), $t(237.73)=0.24$, $p = .810$, $d=0.03 [-0.22, 0.28]$.

Statistical basis: Independent-samples t-test (pooled variance) (Student (1908). "The probable error of a mean." *Biometrika*, 6(1), 1-25.); Welch's correction (unequal variances, the app's default) (Welch, B. L. (1947). "The generalization of Student's problem when several different population variances are involved." *Biometrika*, 34(1-2), 28-35.); Cohen's d effect-size benchmarks (.2/.5/.8) (Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Routledge.); Levene's test for equal variances (Brown-Forsythe median-centered variant) (Levene, H. (1960). "Robust tests for equality of variances." In I. Olkin (Ed.), *Contributions to Probability and Statistics* (pp. 278-292). Stanford University Press.); Brown-Forsythe median-centered variant of the homogeneity-of-variance test (Brown, M. B., Forsythe, A. B. (1974). "Robust tests for the equality of variances." *Journal of the American Statistical Association*, 69(346), 364-367.). Full references in the exported CITATIONS.txt.

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